

Clinical impact of pre-transplant gut microbial diversity on outcomes of allogeneic hematopoietic stem cell transplantation

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Abstract Post-transplant microbial diversity in the gastrointestinal tract is closely associated with clinical outcomes following allogeneic hematopoietic stem cell transplantation (allo-HSCT). However, little is known about the impact of the fecal microbiota before allo-HSCT. We analyzed fecal samples approximately 2 weeks before conditioning among 107 allo-HSCT recipients between 2013 and 2015. Microbial analysis was performed using 16S rRNA gene sequencing. Operational taxonomic unit-based microbial diversity was estimated by calculating the Shannon index. Patients were classified into three groups based on the diversity index: low (<2), intermediate (2, 3), and high (>3) diversity (18 (16.8%), 48 (44.9%), and 41 (38.3%) patients, respectively). There were no significant differences in the 20-month overall survival, cumulative incidence of relapse, and non-relapse mortality among three groups. The cumulative incidence of grade II to IV acute graft-versus-host disease (aGVHD) was similar

among the three groups (low 55.6%; intermediate 35.4%; high 48.8%, $p = 0.339$, at day 100). Furthermore, we found no differences in the cumulative incidence of grade II to IV acute gastrointestinal GVHD among the three groups (low 38.9%; intermediate 21.3%; high 24.4%, $p = 0.778$, at day 100). Regarding the composition of microbiota before allo-HSCT, aGVHD patients showed a significantly higher abundance of phylum *Firmicutes* ($p < 0.01$) and a lower tendency for *Bacteroidetes* ($p = 0.106$) than non-aGVHD patients. Maintenance of *Bacteroidetes* throughout allo-HSCT may be a strategy to prevent aGVHD.

Keywords Intestinal microbiota · Microbial diversity · Allogeneic hematopoietic stem cell transplantation (allo-HSCT) · Acute gastrointestinal graft-versus-host disease (GVHD)

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Introduction

Alterations in the composition of intestinal microbiota are linked to multiple metabolic and inflammatory diseases in humans [1–4]. Recent studies have reported important relationships between the gut microbiota and outcomes in allogeneic stem cell transplantation (allo-HSCT) recipients [5–7]. However, these studies mainly evaluated the fecal bacterial composition after allo-HSCT and did not analyze whether pre-transplant intestinal microbiota would affect transplant outcomes. To clarify this point, we analyzed intestinal microbiota immediately (2 weeks) before allo-HSCT in 107 adult recipients and evaluated the association between pre-transplant microbial diversity and transplant outcomes.

Patients and methods

Patients

This study was approved by the Ethics Committee of Tokyo Metropolitan Cancer and Infectious Diseases Center, Komagome Hospital. Informed consent was obtained from all subjects prior to specimen collection. Subjects consisted of adult patients undergoing allo-HSCT in our institution between April 2013 and March 2015. We excluded patients who had received a previous allo-HSCT, those who did not achieve engraftment after allo-HSCT, and those whose fecal material was insufficient for bacterial DNA extraction.

Transplant procedures

Transplant procedures have been described in detail elsewhere [8]. Generally, patients with lymphoid malignancy were conditioned using a total body irradiation (TBI) regimen (12 Gy) and cyclophosphamide at 120 mg/kg. Patients with myeloid malignancy were conditioned using a non-TBI regimen that included intravenous busulfan at 12.8 mg/kg and cyclophosphamide at 120 mg/kg. The preparative regimen for the reduced-intensity procedure consisted of fludarabine (30 mg/m² for 6 days), busulfan (3.2 mg/kg for 2 days), or melphalan (40 mg/m² for 2 days), and TBI of 400 cGy in divided doses. The patients were given intravenous infusion of donor hematopoietic stem cells on day 0. All patients received acute graft-versus-host disease (aGVHD) prophylaxis with cyclosporine or tacrolimus as well as short-term methotrexate. Tacrolimus was used in cases involving either unrelated or human leukocyte antigen (HLA)-mismatched transplantation. aGVHD was diagnosed and graded according to previously

established criteria [9]. Late-onset aGVHD, occurring more than 100 days after transplantation, was included in the definition of aGVHD [10]. HLA disparities were categorized as either GVHD or rejection direction. The engraftment day was defined as the first day of three consecutive days when the absolute neutrophil count was more than $0.5 \times 10^9/l$. Levofloxacin and itraconazole oral solution were administered 14 days before allo-HSCT for prophylaxis against bacterial and fungal infection, respectively. Trimethoprim and sulfamethoxazole were administered to prevent *Pneumocystis pneumonia* from 14 days before allo-HSCT to the day of transplantation.

Sample analysis

Fecal specimens were collected within 2 weeks before the start of pre-transplant conditioning. Fecal specimens from patients were initially stored at 4 °C and then at –80 °C within 24 h after collection. Bacterial DNA was extracted from the patient's feces.

The hypervariable V1–2 region of the 16S rRNA gene was amplified by polymerase chain reaction (PCR) with barcoded 27Fmod and reverse primer 338R, and then an equal amount of purified PCR amplicons was sequenced using the 454 GS FLX Titanium or 454 Junior system (Roche Applied Science, Indianapolis, IN) [11]. Quality filter-passed reads were obtained by removing reads that did not contain both primer sequences, had an average quality value <25, or were possibly chimeric. Of the quality filter-passed reads, 3000 high-quality reads per sample were randomly chosen. After trimming off both primer sequences, the reads were sorted and grouped into operational taxonomic units (OTUs) using the UCLUST algorithm (<http://www.drive5.com/>) with a sequence identity threshold of 96%. Taxonomic assignments of each OTU were made by similarity searching against the publically available 16S (RDP ver. 10.27 and CORE) and NCBI genome database using the GLSEARCH program. OTU-based microbial diversity was estimated by calculating the Shannon diversity index. Patients were classified into three groups based on the diversity index: low (<2), intermediate (2, 3), and high (>3) diversity groups.

Statistical analysis

The probabilities of overall survival (OS) and disease-free survival were estimated using the Kaplan-Meier product limit method. The cumulative incidences of non-relapse mortality (NRM), relapse, and aGVHD were evaluated using Gray's method, considering each risk as

Table 1 Analysis of fecal samples from 107 patients

Factors		No. of patients			<i>p</i> value	
		High (n=41)	Int (n=48)	Low (n=18)		
Age	<35	5	8	3	0.315	
	35–55	15	25	10		
	>55	21	15	15		
Gender	Female	14	19	10	0.3068	
	Male	27	29	8		
Primary disease	ALL	9	7	1	0.0455	
	AML	11	26	12		
	MDS	14	7	2		
	Others	7	8	3		
Disease risk	CR1	15	23	6	0.4605	
	Non-CR1	26	25	12		
Conditioning	MAC	24	31	10	0.7766	
	RIC	17	17	8		
Stem cell source	CB	3	8	1	0.2988	
	rBM	1	1	0		
	rPBSC	10	10	9		
	uBM	26	29	8		
	uPBSC	1	0	0		
HLA-mismatch (X/6)	0	23	22	9	0.9669	
	1	11	14	4		
	2	3	5	2		
	≥3	4	7	3		
Abx at the collection of fecal samples	Abx	Yes	21	36	16	0.0067
		No	20	12	2	
	Abx effective against GNR		21	36	16	0.369
	Abx effective against anaerobic bacteria		2	4	4	
	Abx non-effective against anaerobic bacteria		19	32	12	0.0195
	Abx effective against MRSA					
	Yes		2	2	5	
	No		39	46	13	
	ST	Yes	22	21	10	0.5603
		No	19	27	8	
HCT-CI	0	17	20	9	0.979	
	1, 2	9	11	3		
	≥3	15	17	6		

ALL acute lymphoid leukemia, AML acute myeloid leukemia, CML chronic myelogenous leukemia, MDS myelodysplastic syndrome, CR1 first complete remission, MAC myeloablative conditioning, RIC reduced intensity conditioning, CB cord blood stem cell, rBM related bone marrow, rPB related peripheral blood stem cell, uBM unrelated bone marrow, uPBSC unrelated peripheral blood stem cell, HLA human leukocyte antigen, Abx antibiotics, GNR gram-negative rods, MRSA methicillin-resistant *Staphylococcus aureus*, ST sulfamethoxazole–trimethoprim, HCT-CI hematopoietic cell transplant-comorbidity index

a competing risk. For each estimation of the cumulative incidence of an event, death without an event was defined as a competing risk. To evaluate the influence of the development of grade II–IV aGVHD, proportional

hazard modeling was used, and the onset of aGVHD was analyzed as a time-dependent covariate. Fisher's exact test was used for categorical variables, and the *t* test was used for continuous variables. All statistical

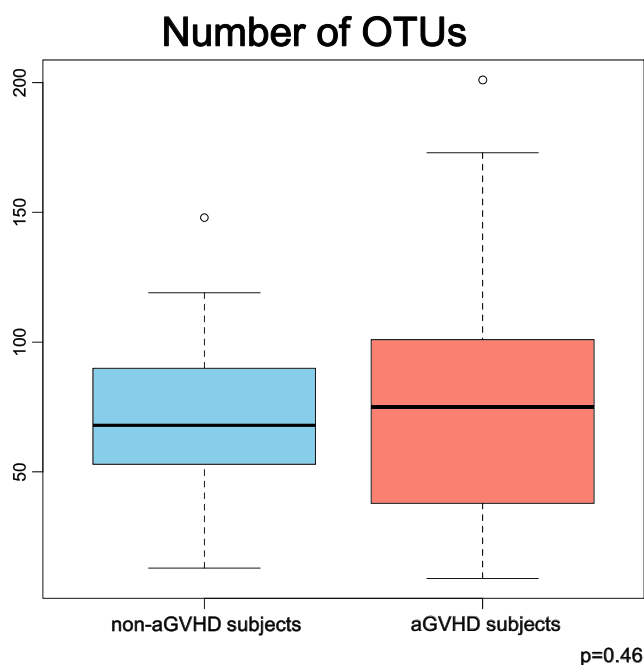
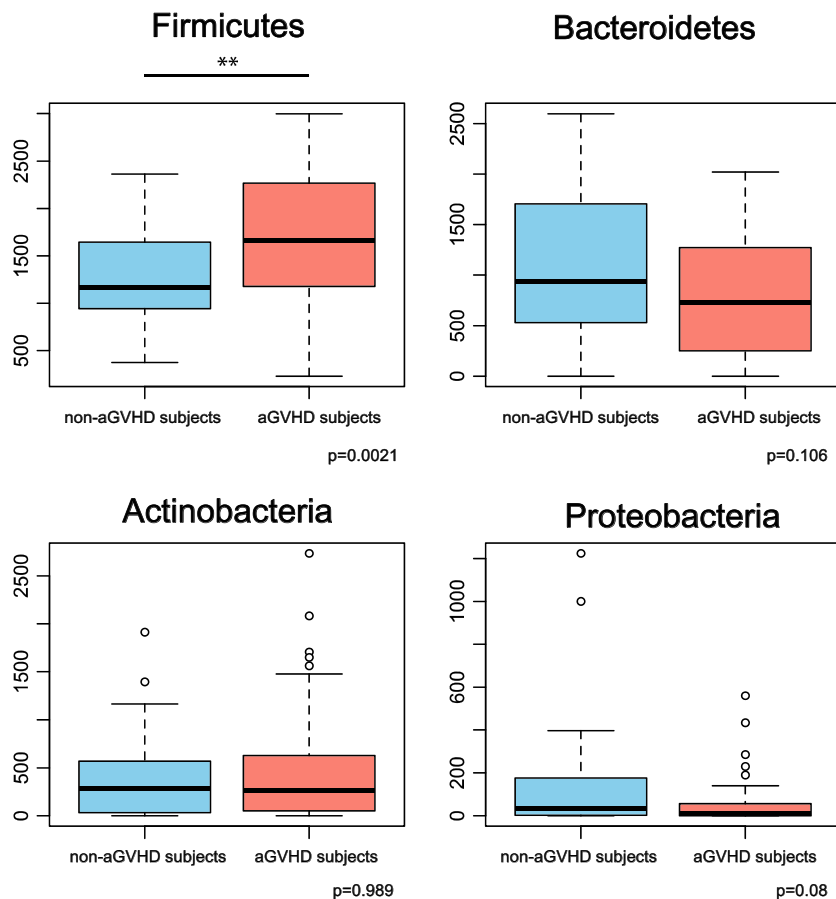


Fig. 1 The number of operational taxonomic units (OTUs) before allogeneic stem cell transplantation (allo-HSCT) is shown as *box plots* for the non-acute graft-versus-host disease (aGVHD) (*left*) and aGVHD (*right*) subjects

Fig. 2 Phylum-level gut microbiota before allo-HSCT. For non-aGVHD (*left*) and aGVHD (*right*) subjects, *box plots* showing the relative abundance of *Firmicutes*, *Bacteroidetes*, *Actinobacteria*, and *Proteobacteria* are shown



analyses were performed with EZR, a graphical user interface for R software (ver. 2.13.0, The R Foundation for Statistical Computing).

Results

Patients' characteristics

Between April 2013 and March 2015, 127 patients eventually underwent their first allo-HSCT in our institution. We excluded 10 patients from whom fecal samples were not obtained, five patients in whom DNA from fecal samples was not extracted, one patient who did not achieve engraftment, and four patients who died from relapse without aGVHD within 100 days after allo-HSCT. Therefore, fecal samples from 107 patients were analyzed.

Among the 107 patients, 18 (16.8%) patients were in the low diversity group, 48 (44.9%) patients were in the intermediate diversity group, and 41 (38.3%) patients were in the high diversity group (Table 1). The median age of all patients was 49 years (range 16–72 years). The median follow-up period for survivors was 597 days (range, 23–1046 days). We found no significant differences among the three groups in terms of

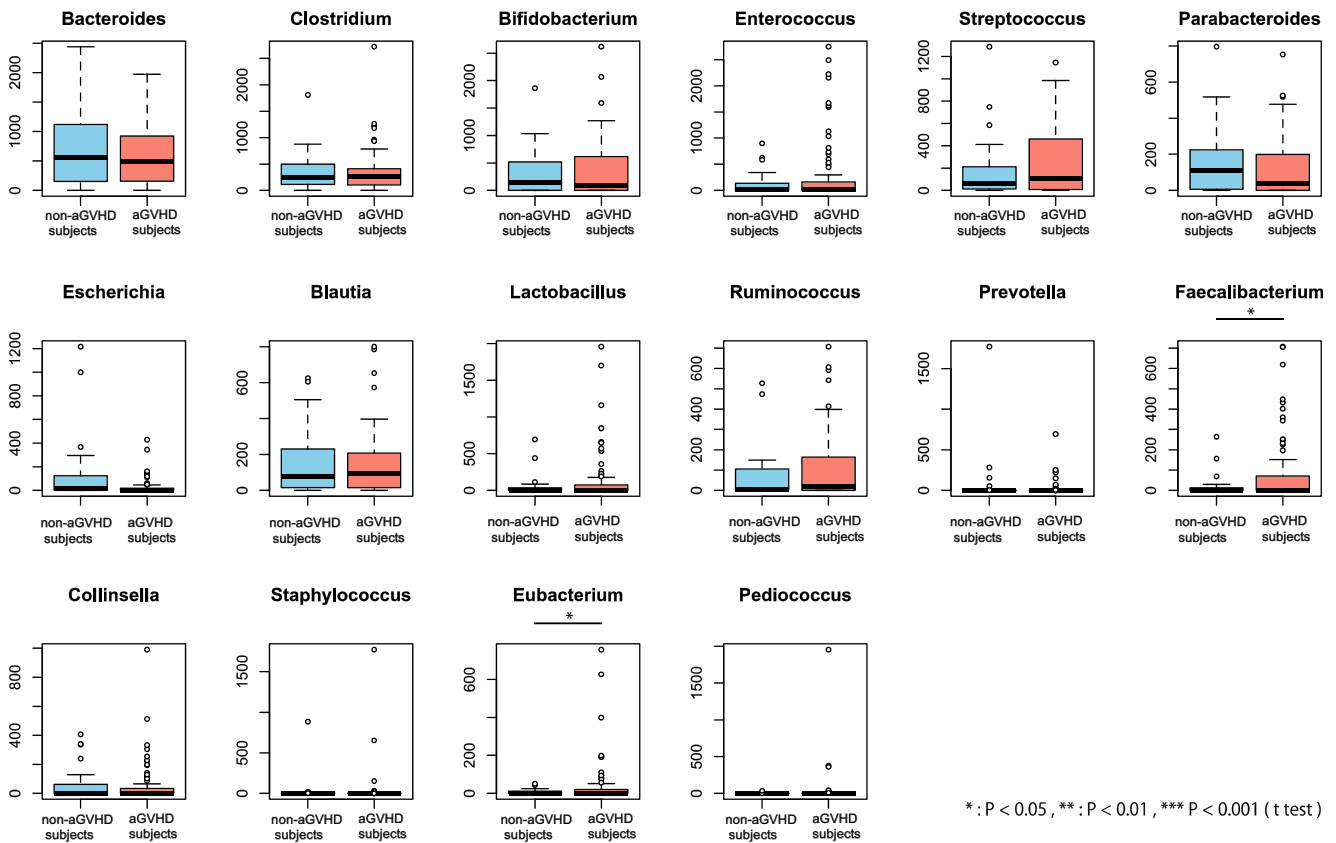


Fig. 3 Genus-level gut microbiota before allo-HSCT. For non-aGVHD (left) and aGVHD (right) subjects, box plots showing the relative abundance of 16 genera are shown

age, sex, disease status, performance status, hematopoietic cell transplantation comorbidity index [12], GVHD prophylaxis, conditioning regimen, stem cell sources, or HLA disparity. Regarding primary diseases, the low diversity group included more acute myeloid leukemia patients (66.7%), and the high diversity group included more acute lymphoid leukemia patients (22%) and myelodysplastic syndrome patients (34.1%). Furthermore, more patients in the low diversity group received intravenous antibiotics before the conditioning regimen than the patients in the intermediate and high diversity groups.

16S rRNA gene sequencing

The OTU counts before allo-HSCT did not differ between the patients with and without aGVHD ($p = 0.46$, Fig. 1). We compared the composition of microbiota before allo-HSCT at the phylum and genus levels between the patients who developed aGVHD and those who did not (Figs. 2 and 3). The pre-allo-HSCT samples from aGVHD patients showed higher abundance of phylum *Firmicutes* compared to those from non-aGVHD samples ($p = 0.0021$, Fig. 2).

On the other hand, we found no statistically significant differences in other phyla such as *Bacteroidetes*, *Actinobacteria*, and *Proteobacteria* between the two groups. However, pre-allo-HSCT samples from aGVHD patients tended to show a lower abundance of phylum *Bacteroidetes* compared to samples from non-aGVHD samples ($p = 0.106$, Fig. 2).

Moreover, the microbial composition, for which we detected more than 100 bacteria at the genus level, showed significant differences only for *Faecalibacterium* and *Eubacterium* ($p < 0.05$) (Fig. 3). However, the number of these bacteria was very small, and the clinical interpretation was unknown.

Transplant outcomes

At 20 months after allo-HSCT, OS for the low diversity group was 61.1% and was similar to those in the intermediate and high diversity groups (intermediate 66.7%; high 73.2%, $p = 0.376$, Fig. 4a). The estimated cumulative incidence of NRM in the low diversity group was 22.2% at 20 months. There were no significant differences among three groups (intermediate 14.6%; high 17.2%, $p = 0.259$, Fig. 4b). We found

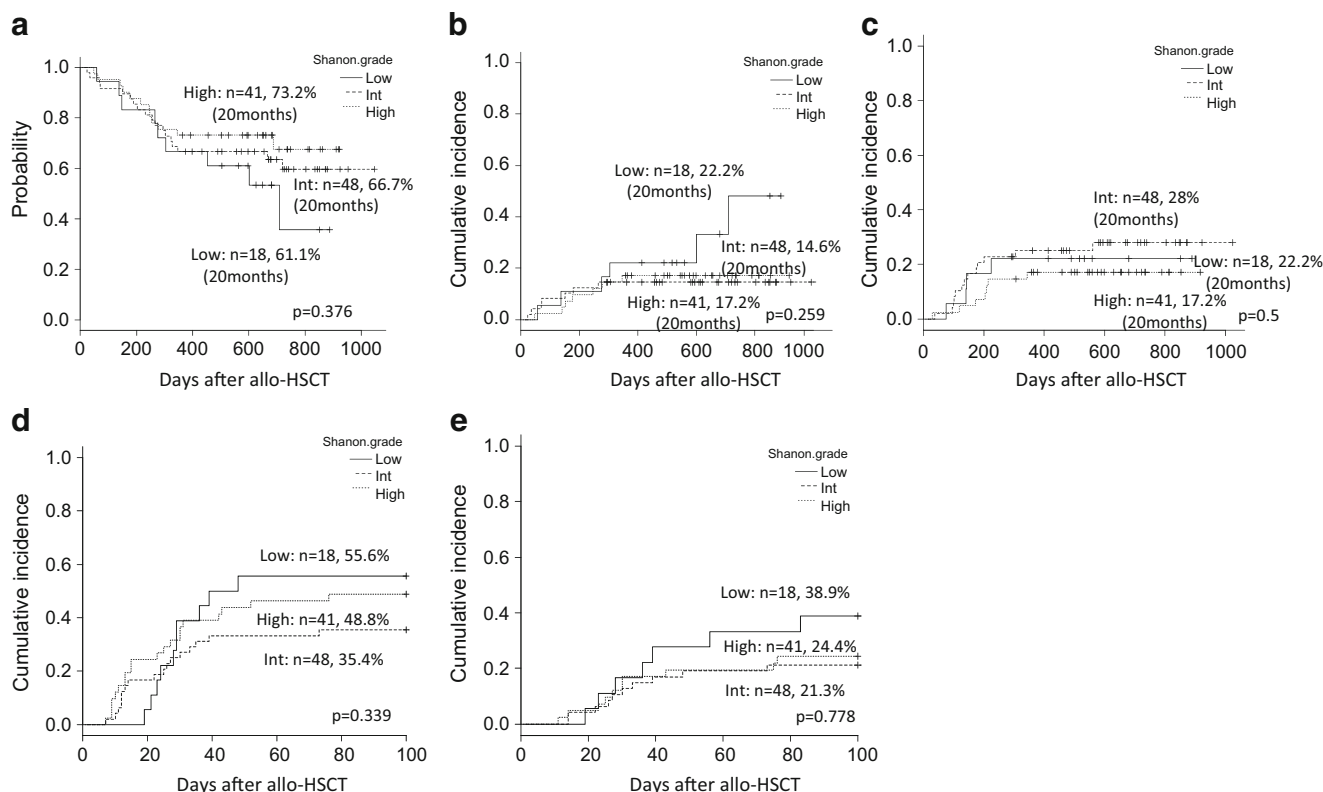


Fig. 4 Overall survival after allogeneic stem cell transplantation (allo-HSCT) in 18 patients in the low diversity group, 48 patients in the intermediate diversity group, and 41 patients in the high diversity group (a). Cumulative incidence of non-relapse mortality after allo-HSCT in the three groups (b). Cumulative incidence of relapse after allo-HSCT in

the three groups (c). Cumulative incidence of grade II to IV acute graft-versus-host disease (GVHD) (d) and grade II to IV acute gastrointestinal GVHD (e) in 18 patients in the low diversity group, 48 patients in the intermediate diversity group, and 41 patients in the high diversity group

no differences in the cumulative incidence of relapse among the three groups (low 22.2%; intermediate 28%; high 17.2%, $p = 0.5$, at 20 months, Fig. 4c). The cumulative incidence of grade II to IV aGVHD was similar among the three groups (low 55.6%; intermediate 35.4%; high 48.8%, $p = 0.339$, at day 100, Fig. 4d). Moreover, there were no significant differences in the cumulative incidence of grade II to IV acute gastrointestinal GVHD at day 100 after allo-HSCT among the three groups (low 38.9%; intermediate 21.3%; high 24.4%, $p = 0.778$, at day 100, Fig. 4e). There were also no significant differences in the cumulative incidence of grade III to IV acute gastrointestinal GVHD at day 100 after allo-HSCT among the three groups (low 19.8%; intermediate 2.3%; high 11.0%, $p = 0.108$, at day 100, data not shown).

Discussion

Taur et al. reported that intestinal microbial diversity at the time of engraftment is associated with mortality, and the low diversity group showed a worse outcome compared to intermediate and high diversity groups [5]. Furthermore, in patients undergoing allo-HSCT, Weber et al. investigated 3-

indoxyl sulfate, which is produced following degradation of tryptophan to indole by intestinal microbiota, followed by microsomal oxidation to indoxyl and sulfonation in the liver [6, 13]. They showed that low urinary 3-indoxyl sulfate levels early after allo-HSCT reflect microbiota disruption and are associated with high transplant-related mortality. Our results demonstrated that low microbial diversity before allo-HSCT was not associated with transplant outcomes including acute gastrointestinal GVHD. These results indicated that intestinal microbial changes early after allo-HSCT may affect transplant outcomes more than pre-transplant changes.

Regarding pre-transplant microbial composition in allo-HSCT recipients, Biagi et al. [14] reported that pre-transplant fecal samples from aGVHD patients demonstrated lower abundance of two members of the phylum *Bacteroidetes* (*Bacteroides* and *Parabacteroides*) and higher abundance of phylum *Firmicutes*, compared to non-aGVHD patients. Chiusolo et al. investigated only eight recipients throughout allo-HSCT [15] and reported that aGVHD patients harbored more *Firmicutes* and *Proteobacteria* and less *Bacteroidetes* than non-aGVHD patients. In our study, we found that pre-transplant fecal samples from aGVHD patients showed a significantly higher abundance of phylum

Firmicutes and a tendency for a lower abundance of *Bacteroidetes* than non-aGVHD patients. These results, including previous reports, indicated that patients who harbor less *Bacteroidetes* before allo-HSCT may have a greater risk of developing aGVHD. However, we should consider that results of microbiota analyses depend on the population background due to differences in nutrition, etc. To clarify these differences, multicenter comparisons are highly needed.

In conclusion, the present study showed that low microbial diversity before allo-HSCT was not associated with transplant outcomes. Bacterial translocation after allo-HSCT is a key trigger for the development of aGVHD. The microbial status early after allo-HSCT may be more important for predicting transplant outcomes than the status before allo-HSCT. Furthermore, non-aGVHD patients harbored more *Bacteroidetes* before allo-HSCT, and thus, maintaining higher levels of *Bacteroidetes* throughout allo-HSCT may be a strategy to prevent aGVHD. Further evaluation with a larger series of patients will be necessary to confirm the association between the pre-transplant microbial ecosystem and transplant outcomes.

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Compliance with ethical standards This study was approved by the Ethics Committee of Tokyo Metropolitan Cancer and Infectious Diseases Center, Komagome Hospital.

Conflict of interest The authors declare that they have no conflict of interest.

Informed consent Informed consent was obtained from all patients prior to being included in the study.

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